

COCSO | Week 2

Obtaining Data

Brendan Tomoschuk

Exam 1 — Monday October 19

- Format: 35 multiple-choice questions, 50 mins, ***CLOSED-NOTE***
 - ***Questions taken from***
 - ***lectures up to and including “Wrangling data”***
 - ***Readings 1-3***
- How to study:
 - Review concepts and clicker questions from each lecture + main ideas/ takeaways from each reading
 - TAs will hold **exam review** during section on **Week 3**
 - Review slides / practice questions will be posted on course website

Guest Lecture — Friday, October 23, 2026

Data Exploration & Startup life

- Participation is worth **1 point**
 - Fill out a Google Form of your takeaways
 - Interactive **Q&A!**
- Dewey data organizes and sells datasets to academic researchers
- Questions examples:
 - Working at at a small company?
 - What kind of data do you work with?



Taylor Butler

Research & Growth Lead

Reading(s) 3: *Data Organization*

Data Organization in Spreadsheets

Karl W. Broman^a and Kara H. Woo^b

^aDepartment of Biostatistics & Medical Informatics, University of Wisconsin-Madison, Madison, WI; ^bInformation School, University of Washington, Seattle, WA

ABSTRACT

Spreadsheets are widely used software tools for data entry, storage, analysis, and visualization. Focusing on the data entry and storage aspects, this article offers practical recommendations for organizing spreadsheet data to reduce errors and ease later analyses. The basic principles are: be consistent, write dates like YYYY-MM-DD, do not leave any cells empty, put just one thing in a cell, organize the data as a single rectangle (with subjects as rows and variables as columns, and with a single header row), create a data dictionary, do not include calculations in the raw data files, do not use font color or highlighting as data, choose good names for things, make backups, use data validation to avoid data entry errors, and save the data in plain text files.

ARTICLE HISTORY

Received June 2017
Revised August 2017

KEYWORDS

Data management; Data organization; Microsoft Excel; Spreadsheets

Tidy Data

Hadley Wickham
RStudio

Abstract

A huge amount of effort is spent cleaning data to get it ready for analysis, but there has been little research on how to make data cleaning as easy and effective as possible. This paper tackles a small, but important, component of data cleaning: data tidying. Tidy datasets are easy to manipulate, model and visualise, and have a specific structure: each variable is a column, each observation is a row, and each type of observational unit is a table. This framework makes it easy to tidy messy datasets because only a small set of tools are needed to deal with a wide range of un-tidy datasets. This structure also makes it easier to develop tidy tools for data analysis, tools that both input and output tidy datasets. The advantages of a consistent data structure and matching tools are demonstrated with a case study free from mundane data manipulation chores.

Keywords: data cleaning, data tidying, relational databases, R.

Theme:
**how to structure and
store data so it's easy to
understand and analyze!**

Reading(s) 3: *Data Organization*

- Readings can be dense and overwhelming—part of this course is learning how to read academic papers!
- Use the Reading Guide to focus on the key ideas
- Reading 3 Quiz: **due Wednesday week 4 @ 11:59pm!**
- 10 MC questions
- 60 mins to complete it
- You may use the reading & reading guide!

R3: Reading Guide

1. [Wickham H, Tidy Data](#)
2. [Woo K & Broman K, Data in Spreadsheets](#)

Why we're reading these

These two readings explain how to structure and store data so it's easy to understand and analyze. Wickham introduces a data-storing standard called "tidy data," while Woo & Broman give practical spreadsheet rules that help you create tidy, reliable data!

Data organization and wrangling are some of the most important (but not always the most glamorous!) parts of the data science process. Taking the time to build your dataset in a tidy manner will save you time (and headaches) later!

How to read these papers

"Tidy Data"

What to look for as you read

1. What exactly counts as a variable, an observation, and a type of observational unit?
2. What are the three rules of tidy data, and what problems do they prevent?
3. What are the common ways data becomes "messy", and what operation fixes each (e.g., melt/reshape, split, cast, separate tables)?
4. Why does Wickham argue that tidy data makes manipulation, visualization, and modeling

Reminder: How to get data

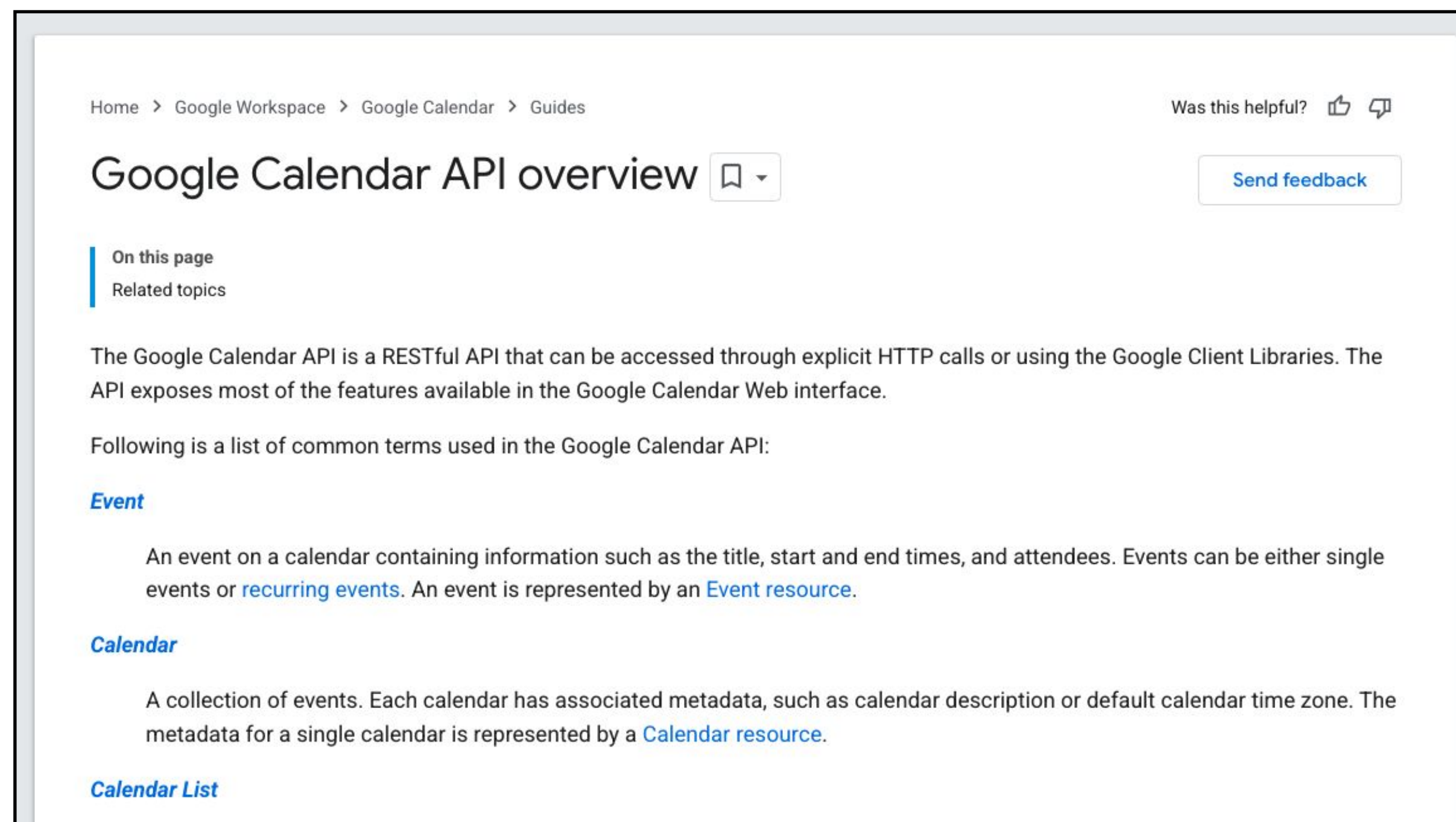
- 1. Collect it yourself
- 2. Use public / open-source data
- 3. Buy it
- 4. Request it from an API
- 5. Scrape it from a website
- 6. Source it from your company

Primary Vs. Secondary Data Analysis	
Primary	Secondary
Data is collected first hand for a specific research aim.	Data was collected by someone else for a different purpose
Data is up-to-date as it is collected for the current research.	Data may be outdated and not specific to current research.
Expensive and time consuming to collect.	Less expensive and saves time as data already exists.
Requires ethical considerations such as informed consent.	Data is in the public domain and has fewer ethical concerns.
Researcher has control over data and accuracy.	Researcher has no control over data accuracy.

Consider the pros and cons of **collecting data yourself (primary)** vs. **getting it from somewhere else (secondary)**

4. Requesting it from an API

- API stands for [Application Programming Interface](#)
 - A way for software programs to talk to each other
- **Example:** Where does all of my time go in one week?!...
- I can collect data from my own calendar using the [Google Calendar API](#)



4. Requesting it from an API

Collecting data using an API

1. Send a **request** (e.g., to retrieve my calendar events)
2. API sends back **semi-structured data**, often JSON (e.g. calendar event start times/end times)

```
{
  "summary": "Prep COGS 9 lecture",
  "start": { "dateTime": "2026-02-02T09:00:00-08:00" },
  "end": { "dateTime": "2026-02-02T11:00:00-08:00" }
},
{
  "summary": "Gym",
  "start": { "dateTime": "2026-02-02T17:30:00-08:00" },
  "end": { "dateTime": "2026-02-02T18:30:00-08:00" }
},
{
  "summary": "Meeting with TA team",
  "start": { "dateTime": "2026-02-03T13:00:00-08:00" },
  "end": { "dateTime": "2026-02-03T14:00:00-08:00" }
},
}
```

{.json}

JSON (JavaScript Object Notation)

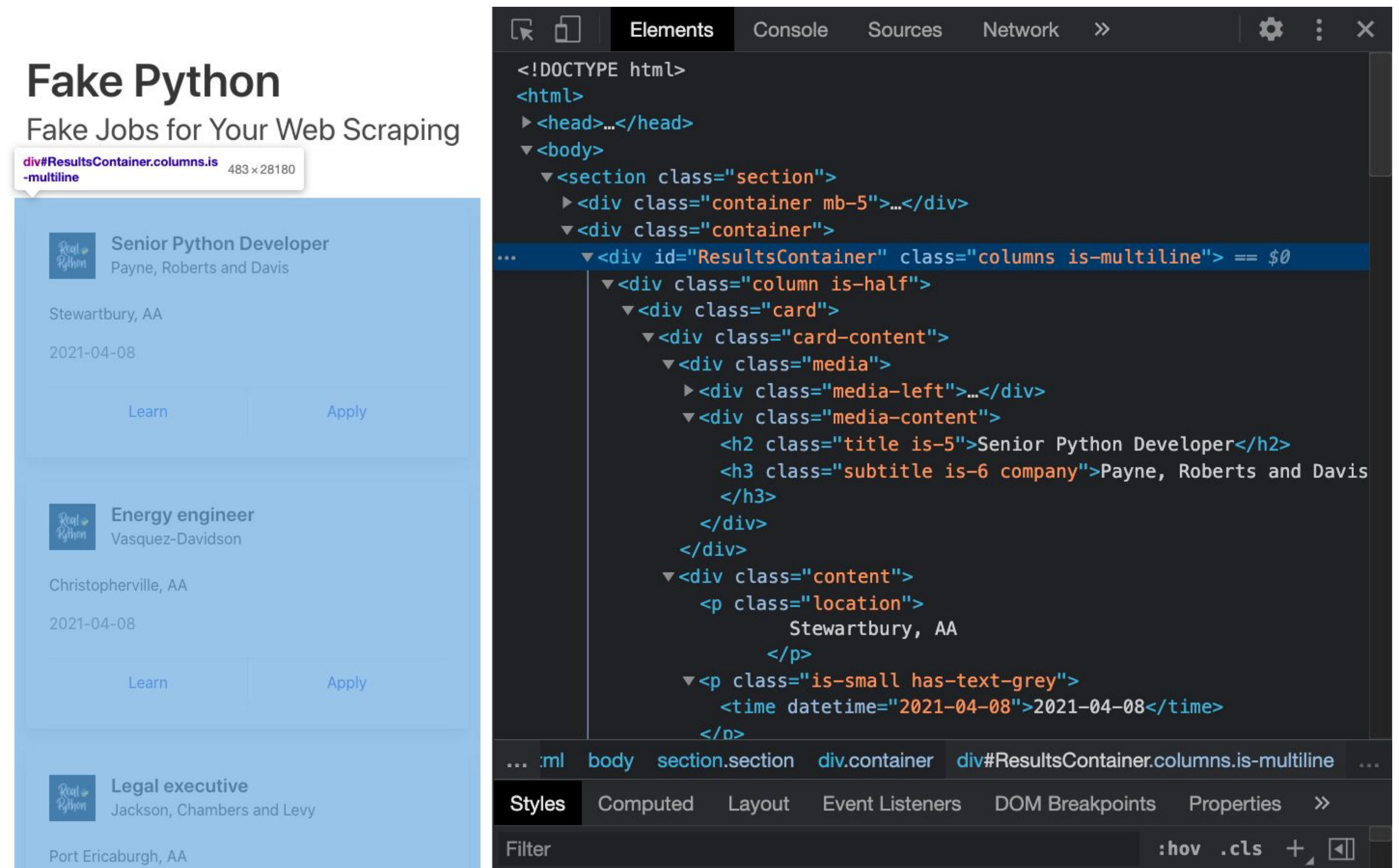
4. Requesting it from an API

Common APIs

- **Maps & locations:** Google Maps
- **Weather:** OpenWeather, WeatherAPI, NOAA (US)
- **Video & music:** YouTube Data API, Spotify Web API
- **Payments:** Stripe API, PayPal API
- **Auth / login:** “Sign in with Google” (Google OAuth)

5. Web scraping

- A way to automatically collect information from websites
- You write a script that downloads a webpage and extracts specific information from the raw HTML code
- Used when there's no API or the API doesn't provide the data you need
- **Example:** You want to build a dataset of entry-level data science jobs



The image displays a web page titled "Fake Python" with the subtitle "Fake Jobs for Your Web Scraping". The page lists three job entries, each with a "Real Python" logo, a job title, company name, location, date, and "Learn" and "Apply" buttons.

The job listings are:

- Senior Python Developer** by Payne, Roberts and Davis, located in Stewartbury, AA, dated 2021-04-08.
- Energy engineer** by Vasquez-Davidson, located in Christopherville, AA, dated 2021-04-08.
- Legal executive** by Jackson, Chambers and Levy, located in Port Ericaburgh, AA.

On the right, the browser's developer tools are open to the "Elements" tab, showing the HTML structure. The selected element is a `div` with the ID `ResultsContainer` and class `columns is-multiline`. The structure shows a `section` containing a `div` with class `container mb-5`, which in turn contains a `div` with class `container`. This container holds a `div` with class `column is-half`, which contains a `div` with class `card`. The card contains a `div` with class `card-content`, which includes a `div` with class `media`. The media section contains a `div` with class `media-left` and a `div` with class `media-content`. The `media-content` div contains an `h2` with class `title is-5` (Senior Python Developer), an `h3` with class `subtitle is-6 company` (Payne, Roberts and Davis), and a `div` with class `content`. The `content` div contains a `p` with class `location` (Stewartbury, AA) and a `p` with class `is-small has-text-grey` (2021-04-08).

5. Web scraping

- **Example:** You want to build a dataset of entry-level data science jobs
 - Check job description for degree requirements (e.g., no “Master’s degree”)
 - When was the job posted?
 - Find the link to apply

HTML

```
<!-- ... -->
<footer class="card-footer">
  <a href="https://www.realpython.com" target="_blank"
    class="card-footer-item">Learn</a>
  <a href="https://realpython.github.io/fake-jobs/jobs/senior-python-dev"
    target="_blank"
    class="card-footer-item">Apply</a>
</footer>
</div>
</div>
```

Python

```
>>> for job_card in python_job_cards:
...     links = job_card.find_all("a")
...     for link in links:
...         link_url = link["href"]
...         print(f"Apply here: {link_url}\n")
... 
```

Python

scraper.py

```
import requests
from bs4 import BeautifulSoup

URL = "https://realpython.github.io/fake-jobs/"
page = requests.get(URL)

soup = BeautifulSoup(page.content, "html.parser")
results = soup.find(id="ResultsContainer")

python_jobs = results.find_all(
    "h2", string=lambda text: "python" in text.lower()
)

python_job_cards = [
    h2_element.parent.parent.parent for h2_element in python_jobs
]

for job_card in python_job_cards:
    title_element = job_card.find("h2", class_="title")
    company_element = job_card.find("h3", class_="company")
    location_element = job_card.find("p", class_="location")
    print(title_element.text.strip())
    print(company_element.text.strip())
    print(location_element.text.strip())
    link_url = job_card.find_all("a")[1]["href"]
    print(f"Apply here: {link_url}\n")
```


Web scraping vs. APIs

- **APIs:** a site's official way to access data
 - structured and stable
 - may require a key & have rate limits
- **Web scraping:** useful when there's no API
 - more fragile (websites change)
 - can violate terms & conditions
- **Takeaway:** Use an API when available; scrape only when necessary and responsibly!

Criteria	Web Scraping	API
Availability	High availability, as web scraping can be applied to virtually any website	Limited availability, with less than 1% of websites offering an API
Data Attributes	Highly flexible, allowing the user to define custom data extraction parameters for publicly available data	Limited control, with the API's structure and data attributes dictated by the provider
Stability	Moderately stable, with reliance on the consistent structure of targeted websites	Generally high stability, as APIs are maintained and controlled by the providers
Technical difficulty	Requires significant technical expertise, especially for complex websites	Relatively easy to use with basic programming skills, simplifying data access
Cost	Cost varies based on different factors; more cost-effective at larger scales	Costs are predictable, though they can escalate significantly at very large scales
Data Quality	Necessitates robust data cleaning systems to ensure quality	Typically offers structured data, often presented in JSON format

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Webclicker

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webclicker.web.app



course code:

WSTAPX

You want to analyze current weather conditions for cities around the world and update them hourly. What is the best way to obtain this data?

- (A) Collect it yourself
- (B) Buy it
- (C) Use public / open-source data
- (D) Request it from an API
- (E) I'm not sure / I'm confused

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What are some things we should know
about our data before using it?

(~2 min with your neighbor)

Questions to ask *before* using data

- Am I allowed to use this data? Are there ethical, privacy, or legal restrictions?
- Why and how was this data collected?
- How was the data cleaned or altered? Is there missing data, and why?
- Who or what is included or excluded? Are there biases in data collection or measurement?
- Is there documentation? (data dictionary, codebook, known issues)
- Can I trust the source?

Unethical web scraping

⚠ Always read the
Terms of Service / User Agreement!

- Scraping behind logins/paywalls without permission
- Scraping personal data (PII) without consent (privacy law risk: GDPR/CCPA)
- Scraping copyrighted/proprietary content without a license



LinkedIn sued ProAPIs and its CEO Rahmat Alam for running millions of fake accounts to scrape and sell user data, charging up to \$15,000 per month.

LinkedIn

User Agreement ▼ Privacy Policy Professional Community Policies Cookie Policy Copyright Policy Regional Info ▼

User Agreement

Effective on November 3, 2025

8.2. Don'ts

You agree that you will *not*:

1. Create a false identity on LinkedIn, misrepresent your identity, create a Member profile for anyone other than yourself (a real person), or use or attempt to use another's account (such as sharing log-in credentials or copying cookies);
2. Develop, support or use software, devices, scripts, robots or any other means or processes (such as crawlers, browser plugins and add-ons or any other technology) to **scrape** or copy the Services, including profiles and other data from the Services;
3. Override any security feature or bypass or circumvent any access controls or use limits of the Services (such as search results, profiles, or videos);

robots.txt

- **robots.txt**: small text file a website can publish that tells automated crawlers (search engine bots, web scrapers) which parts of the site they're **allowed** to access
- It's a **voluntary policy** (bad robots can disobey...)
- Websites enforce real restrictions with logins, paywalls, and server-side blocking



Robots.txt

User-agent: *

Disallow:



Allow:

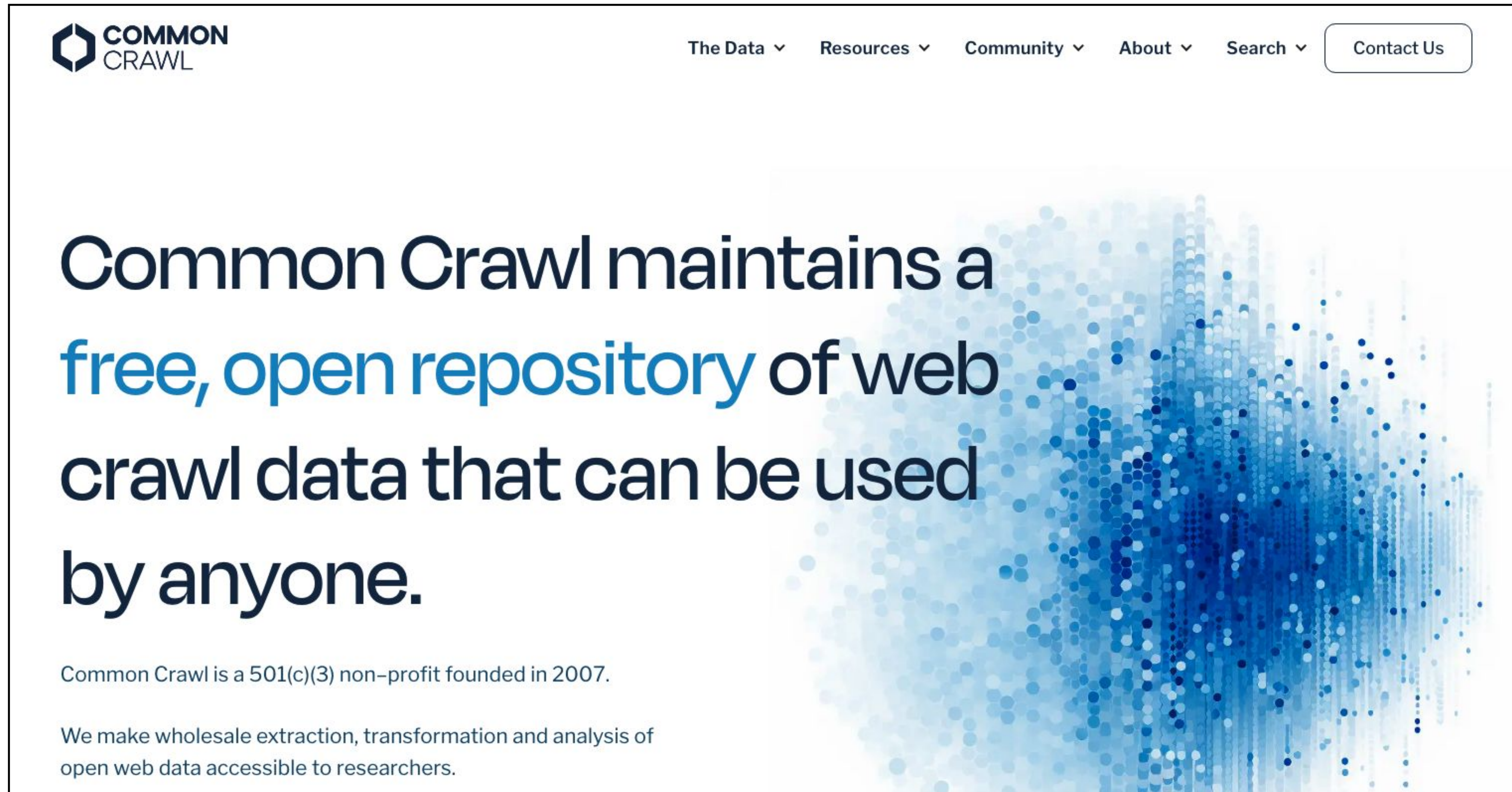


Sitemap:



What about web scraping to train AI models?

An estimated ~60% of GPT-3 was trained on...

The image shows a screenshot of the Common Crawl website. The header features the Common Crawl logo on the left and a navigation menu on the right with links for 'The Data', 'Resources', 'Community', 'About', 'Search', and a 'Contact Us' button. The main content area has a large heading and a background graphic of blue dots of varying sizes. Below the heading, there is a paragraph of text and a small line of text at the bottom left.

COMMON CRAWL

The Data ▾ Resources ▾ Community ▾ About ▾ Search ▾ [Contact Us](#)

Common Crawl maintains a free, open repository of web crawl data that can be used by anyone.

Common Crawl is a 501(c)(3) non-profit founded in 2007.

We make wholesale extraction, transformation and analysis of open web data accessible to researchers.

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.



Share full article



1.3K



KATE KNIBBS

BUSINESS OCT 7, 2024 7:30 AM

The Race to Block OpenAI's Scraping Bots Is Slowing Down

OpenAI's spree of licensing agreements is paying off already—at least in terms of getting publishers to lower their guard.



Webscrapping, APIs and LLMs

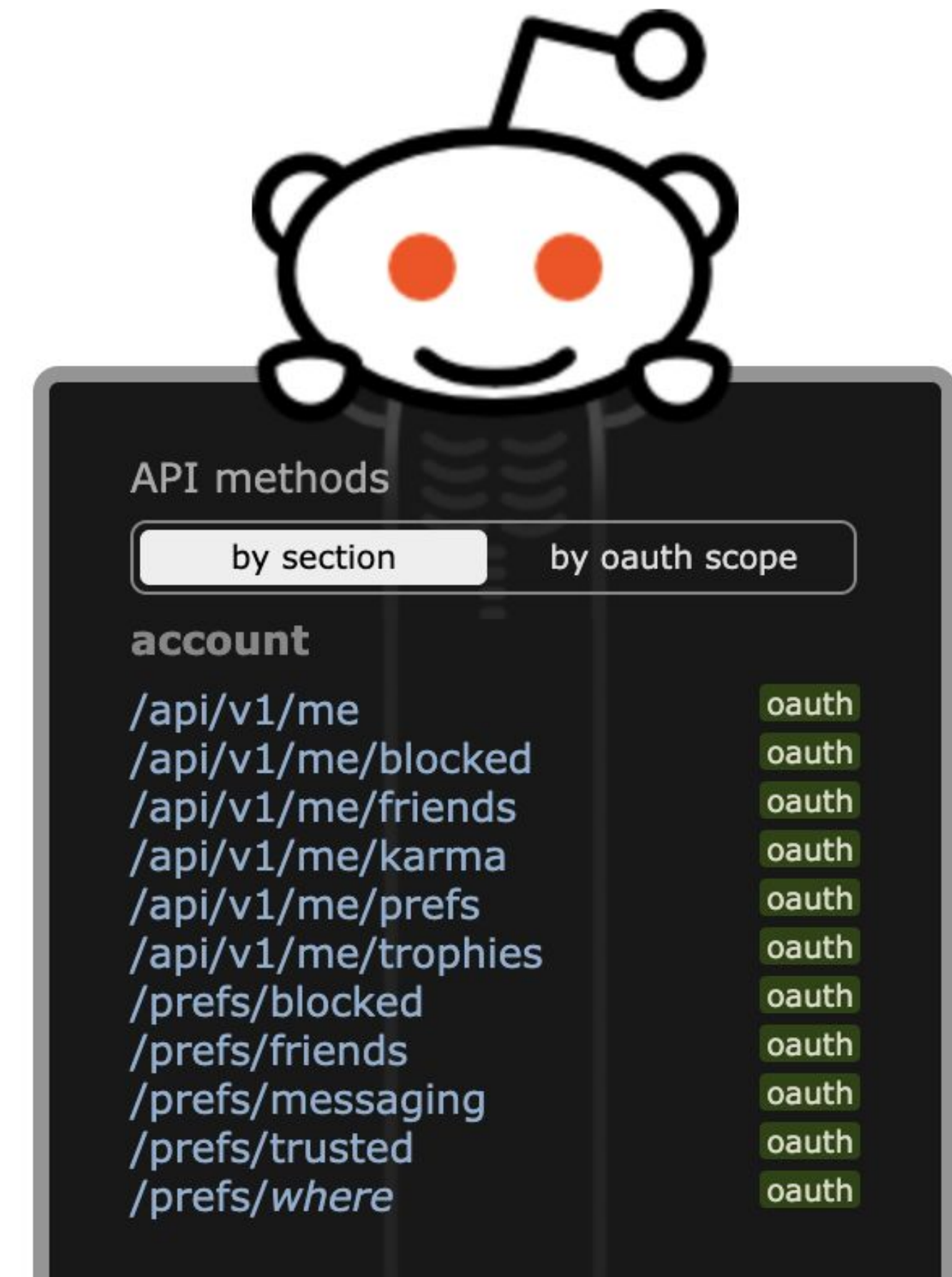
- **Example: Reddit**
 - Pre-LLMs: Free API, no need to web scrape
 - Post-LLMs: Expensive API, web scraping blocked



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Reddit Case Study — The Great API Blackout and How It Led to a \$6.4 Billion IPO

AI data licensing emerged as the business model that justified the pricing decision. In early 2024, Reddit announced a data licensing agreement with **Google** valued at approximately **\$60 million annually**



Some of OpenAI's arguments

- Training LLMs is **highly transformative** and protected by **fair use**
 - **Fair use:** doctrine in U.S. copyright law that allows *limited use of copyrighted material without permission* when the use serves **socially valuable purposes**
- The model's outputs are not NYT's works **verbatim** (no direct copying)
 - OpenAI argues that ChatGPT generates new text *informed* by patterns in training data
- ***What do you think?*** (~5-7 min with your neighbor)

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course code:
WSTAPX

What do you think? OpenAI should be able to train their models on public internet data...

- (A) with no restrictions / under any circumstances
- (B) unless the content is explicitly restricted or opted out
- (C) only with explicit permission from the content owner
- (D) OpenAI should NOT be able to train on any public internet data under any circumstance

Takeaways

- There are **many ways** to obtain data—let your **question** guide your choice of data acquisition
- Ethical and responsible data use is an **integral part of the data science process**
 - Always consider how the data was obtained, any restrictions on its use, and privacy concerns (especially when working with personal data)